

BERLIN TEMPELHOF, Germany

WMO: 103840

Lat: 52.4675N

Lon: 13.4022E

Elev: 161

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 86-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.7	17.0	5.8	7.5	12.6	11.4	9.9	19.3	27.8	44.9	24.9	43.4	7.4	80	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	86.4	66.3	82.7	65.2	79.5	63.9	68.9	80.4	67.4	78.1	65.7	75.8	8.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	94.2	72.3	63.9	89.5	71.4	62.2	84.1	69.9	33.1	81.3	31.9	78.0	30.6	75.5	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.3	19.6	17.5	DB	10.9	93.3	9.5	3.9	4.1	96.1	-1.4	98.4	-6.8	100.6	-13.6	103.5
			WB	10.1	72.3	9.2	1.8	3.5	73.6	-1.9	74.7	-7.1	75.7	-13.8	77.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	34.4	34.4	40.9	48.7	57.6	63.4	67.5	65.5	59.2	50.7	41.4	36.5	(6)
	DBStd	13.83	9.73	8.59	8.06	6.91	7.06	6.71	6.18	5.76	5.35	6.56	6.16	6.97	(7)
	HDD50	2092	483	438	294	104	13	0	0	0	1	74	264	421	(8)
	HDD65	5728	947	856	749	488	247	109	43	66	188	443	708	884	(9)
	CDD50	2133	1	2	11	66	248	402	542	479	278	96	6	2	(10)
	CDD65	294	0	0	0	0	17	61	121	80	15	0	0	0	(11)
	CDH74	2400	0	0	0	9	166	500	1037	574	114	0	0	0	(12)
	CDH80	669	0	0	0	0	28	118	342	172	9	0	0	0	(13)
(14) Wind	WSAvg	8.8	10.5	9.8	9.8	8.5	8.2	7.8	8.1	7.6	8.0	8.2	9.2	9.7	(14)
(15) Precipitation	PrecAvg	22.7	1.8	1.3	1.5	1.4	2.1	2.6	2.5	2.4	1.8	1.6	1.9	1.9	(15)
	PrecMax	32.9	4.7	3.5	4.6	3.5	5.8	7.6	7.3	6.3	5.3	5.7	4.0	5.1	(16)
	PrecMin	14.8	0.2	0.1	0.4	0.0	0.2	0.1	0.3	0.3	0.2	0.2	0.0	0.4	(17)
	PrecStd	4.3	0.9	0.8	0.9	0.8	1.2	1.7	1.4	1.3	1.0	1.1	0.8	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	58.0	64.3	75.1	84.3	87.7	92.2	91.2	81.0	70.8	57.4	55.3	(19)
		MCWB	49.9	50.8	52.9	56.4	65.4	65.7	67.2	69.3	64.6	58.9	52.7	51.1	(20)
	2%	DB	51.4	52.5	59.3	69.7	78.1	83.5	87.5	84.5	77.3	66.3	53.9	51.9	(21)
		MCWB	47.4	47.2	49.7	55.3	61.2	64.6	66.6	66.5	62.5	57.6	50.5	48.1	(22)
	5%	DB	48.2	48.2	55.5	64.8	74.9	79.7	83.9	80.3	73.4	63.5	52.0	49.6	(23)
		MCWB	45.3	44.2	48.1	52.6	59.2	63.4	65.6	64.8	61.4	56.0	49.0	46.5	(24)
	10%	DB	46.0	45.0	52.1	60.7	71.4	76.2	80.3	76.1	69.2	60.6	50.0	46.7	(25)
		MCWB	43.0	41.7	46.1	50.7	57.6	62.2	64.6	63.6	60.1	54.3	47.5	44.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	51.6	54.2	57.9	67.2	69.9	71.5	70.4	67.3	60.8	53.6	51.5	(27)
		MCDB	54.2	57.2	61.6	70.0	78.0	80.7	83.1	85.6	77.8	67.1	56.6	55.0	(28)
	2%	WB	47.9	47.6	51.4	55.9	64.2	67.4	69.2	68.4	64.7	58.6	51.4	48.6	(29)
		MCDB	50.7	51.9	57.4	67.0	73.5	77.6	80.6	79.4	71.9	65.3	53.7	51.5	(30)
	5%	WB	45.3	44.8	49.2	53.7	61.3	65.4	67.6	66.8	62.8	56.7	49.6	46.4	(31)
		MCDB	47.6	47.7	54.2	63.9	71.3	75.4	79.2	76.5	69.9	61.9	51.8	49.1	(32)
	10%	WB	43.0	41.8	46.8	51.5	58.9	63.5	65.7	65.1	60.8	55.1	47.8	44.2	(33)
		MCDB	45.3	44.5	51.7	59.2	69.2	73.8	76.9	73.9	68.8	59.9	49.7	46.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.7	12.3	15.5	17.0	16.1	16.4	15.3	14.5	13.2	8.6	7.4	(35)
		MCDBR	9.1	13.9	18.1	22.4	22.6	21.8	23.0	22.7	22.1	17.3	9.7	9.2	(36)
	5% WB	MCWBR	7.9	9.9	10.8	11.4	10.3	8.8	8.6	9.6	10.1	10.1	7.6	7.5	(37)
		MCWBR	9.0	13.3	16.7	19.6	20.1	19.4	19.5	18.4	17.3	14.9	9.2	9.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.327	0.353	0.394	0.419	0.416	0.413	0.434	0.427	0.389	0.376	0.351	0.321	(40)
		taud	2.310	2.285	2.193	2.197	2.250	2.282	2.236	2.268	2.350	2.346	2.351	2.323	(41)
	Ebn at Noon		207	236	248	258	265	267	258	252	249	225	196	187	(42)
		Edh at Noon	20	27	36	41	41	40	41	38	31	26	20	17	(43)
(44) All-Sky Solar Radiation	RadAvg		200	421	746	1247	1544	1642	1539	1324	966	536	246	149	(44)
	RadStd		22	73	83	161	184	129	204	106	110	64	31	16	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page